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Facultad de Ciencias

GRADO EN FÍSICA

Práctica de Complejos

Quantum Chaos and the Statistics of Energy Levels

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Abstract

In this report, we study how classical chaos shows up in quantum mechanics by looking at the statistical properties of energy spectra in 2D billiards. To do this, we solved the time-independent Schrödinger equation using a finite difference approach and diagonalized the resulting sparse matrices. We analyzed two specific shapes with completely different classical dynamics: a rectangular billiard (which is an integrable system) and a quarter-stadium billiard (which is a chaotic system).

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1 Introduction

The study of complex systems allows to explore the transition between the classical order and the statistichal manifestations in the quantum regime. As in classical mechanics chaos isdefined as the exponential sensitivity to initial conditions, the linearity in the Schrödinger equation does not allow for a direct translation of this definition to quantum mechanics. However, the phenomena known as 'Quantum Chaos' is manifested throughout the statistichal properties of the energy spectrum of the system.

The nuclei in this project is the Bohigas-Giannoni-Schmit (BGS) conjecture, which states that the energy levels of a quantum system whose classical counterpart is chaotic will exhibit the same statistical properties as the eigenvalues of a random matrix ensemble.

To validate this hipothesis, we'll investigate the distribution of spacing between adjacent energy levels in two systems (s_i): one integrable and one chaotic.

The first one is the two-dimensional **rectangular billiard**, which is an **integrable system**. In this system, the random matrix theory (RMT) predicts that the distribution of level spacings follows a Poisson distribution, which is given by:

$$P(s) = e^{-s} \tag{1.1}$$

The second system is the **stadium billiard**, which is a **chaotic system**. In this case, RMT predicts that the distribution of level spacings follows the Wigner-Dyson distribution, which is given by:

$$P(s) = \frac{\pi}{2} s e^{-\frac{\pi}{4}s^2} \tag{1.2}$$

where s is the normalized spacing between adjacent energy levels in both cases.

To integrate the time independent Schrödinger equation for these systems, we will use the **Finite Difference Method** (FDM): this method allows us to discretize the spatial domain and solve for the energy eigenvalues and eigenfunctions of the system.

2 Methods

To be able to solve the Schrödinger's Equation $H\psi = E\psi$, we'll first need to discretize the space and then solve the resulting eigenvalue problem. We'll need to impose the Dirichlet conditions to the domain, which means that the wavefunction ψ must vanish at the boundaries of the billiard.

$$\psi|_{\partial\Omega} = 0 \tag{2.1}$$

All the methods used in this project are implemented in Python, and the code is available in the GitHub repository . To explain the followed procedure, we'll first introduce the Mascara method, which is used to discretize the system, and then the Finite Difference Method, which is used to solve the resulting eigenvalue problem.

2.1 Discretisation of the system: Mascara Method

To easily create different geometries, we decided to implement a finite difference method that takes different masks and computes the corresponding eigenvalues. The mask represents a grid containing the geometry of the problem. For each pixel, we determine whether it belongs to the geometry by assigning it a true or false value.

To carry out this project, we created three different masks: a rectangular mask, a stadium-shaped mask, and a quarter-stadium mask. The following figure shows these masks:

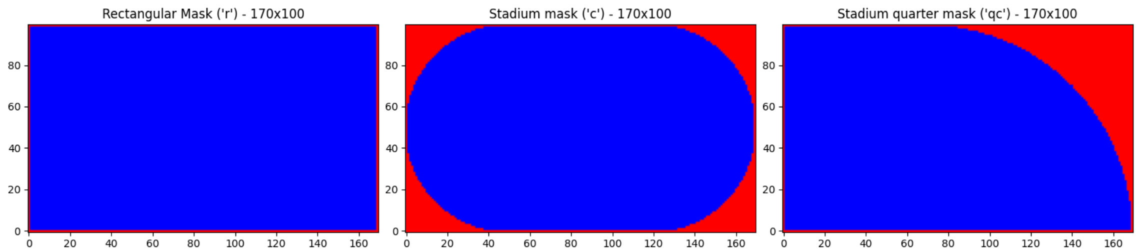


Figura 1: Different mask geometries used in this experiment. From left to right: a rectangular mask, a stadium-shaped mask, and a quarter-stadium mask.

The project aimed to investigate the stadium geometry, which produces chaotic behavior in the eigenvalue spectrum. However, we encountered a problem: the stadium has two symmetries, one with respect to the x-axis and another with respect to the y-axis. In Chapter 1.4 of the thesis by *Wisniacki, Diego A.*, it is explained that the eigenvalues are divided into four different sets because of these symmetries:

- even | even
- even | odd
- odd | even
- odd | odd

This produces four different Wigner–Dyson distributions, which, when combined, result in a Poisson distribution. By removing the symmetries and considering only a quarter of the stadium, the chaotic behavior becomes visible. [1]

2.2 Finite Difference Method

Once we have a grid of the system, where $x_i = i\Delta x$ and $y_j = j\Delta y$, being $i = 1, 2, ...n_x$ and $j = 1, 2, ...n_y$. As we know, the Hamiltonian for a two-dimensional system is given by:

$$H = -\frac{\hbar^2}{2m}\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right) + V(x, y) \tag{2.2}$$

Since in our systems we have a free particle, the potential $V(x,y)$ is zero everywhere inside the billiard and infinite outside. With this consideration, the Schrödinger equation can be rewritten as the Helmholtz equation:

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + k^2\right)\psi(x,y) = 0 \tag{2.3}$$

where $k^2 = E$ (using units such that $\hbar^2/2m = 1$).

Now, we can discretize the second derivatives using finite differences. If we consider one of the dimensions, for example the x-direction, we can approximate the wave function in the nearest points with a Taylor expansion:

$$\psi(x_i \pm \Delta x) = \psi(x_i) \pm \Delta x \psi'(x_i) + \frac{\Delta x^2}{2}\psi''(x_i) \pm O(\Delta x^3) \tag{2.4}$$

Reordenating terms, we can get an expression for the second derivative:

$$\psi''(x_i) = \frac{\psi(x_i + \Delta x) - 2\psi(x_i) + \psi(x_i - \Delta x)}{\Delta x^2} + O(\Delta x^2) \approx \frac{\psi_{i+1} - 2\psi_i + \psi_{i-1}}{\Delta x^2} \tag{2.5}$$

Taking this in consideration, and applying the same procedure for the y-direction, we'll apply the Helmholtz equation in its discretized form 2.6 to the point (x_i, y_j) of the grid.

$$\left(\frac{\psi_{i+1,j} - 2\psi_{i,j} + \psi_{i-1,j}}{\Delta x^2} + \frac{\psi_{i,j+1} - 2\psi_{i,j} + \psi_{i,j-1}}{\Delta y^2} + k^2\psi_{i,j}\right) = 0 \tag{2.6}$$

Is more efficient to write the resulting equation in a matrix form, where the matrix H is given by:

$$H' = \begin{pmatrix} D & \frac{1}{\Delta x^2} & 0 & \dots & \frac{1}{\Delta y^2} & 0 & \dots \\ \frac{1}{\Delta x^2} & D & \frac{1}{\Delta x^2} & 0 & \dots & \frac{1}{\Delta y^2} & \dots \\ 0 & \frac{1}{\Delta x^2} & D & \frac{1}{\Delta x^2} & 0 & \dots & \dots \\ \vdots & 0 & \ddots & \ddots & \ddots & 0 & \vdots \\ \frac{1}{\Delta y^2} & \vdots & 0 & \frac{1}{\Delta x^2} & D & \frac{1}{\Delta x^2} & 0 \\ 0 & \frac{1}{\Delta y^2} & \vdots & \dots & \frac{1}{\Delta x^2} & D & \frac{1}{\Delta x^2} \\ \vdots & \dots & \dots & \dots & 0 & \frac{1}{\Delta x^2} & D \end{pmatrix} \tag{2.7}$$

This matrix is called the finite difference matrix (FDM) and it is a sparse matrix, since most of its elements are zero. It has a total of $n_x \times n_y$ elements. The diagonal elements are given by $D = k^2 - \left(\frac{2}{\Delta x^2} + \frac{2}{\Delta y^2}\right)$. The second diagonal elements are going by $\frac{1}{\Delta x^2}$ and it corresponds to the coupling between the nearest points in the x-direction. The elements that are n_x positions away from the diagonal are given by $\frac{1}{\Delta y^2}$ and it corresponds to the coupling between the nearest points in the y-direction. Since in our systems we have a free particle, the potential $V(x,y)$ is zero everywhere inside the billiard and infinite outside.

2.3 Eigenvalues distribution

Then we calculate the eigenvalues and the eigenvectors of the (2.7), with the help of the library scipy. Now we calculate the difference between consecutive eigenvalues, which is called the spacing distribution (and normalized), and we plot an histogram with those values. This histogram will give us information about the chaocity of the system, since, as we know, the spacing distribution for a chaotic system follows a Wigner-Dyson distribution, while for an integrable system it follows a Poisson distribution.

3 Results

3.1 Statistichal analysis of spacing distribution

First,we calculate the first 10000 eigenvalues of each system by diagonalizing the sparse matrix that we obtained applying the DFM 2.7. Then, with those arrays, we compute a new one that has the spacing distribution between consecutive eigenvalues, this is $s = E_{i+1} - E_i$. Finally, we plot an histogram with those, as it follows:

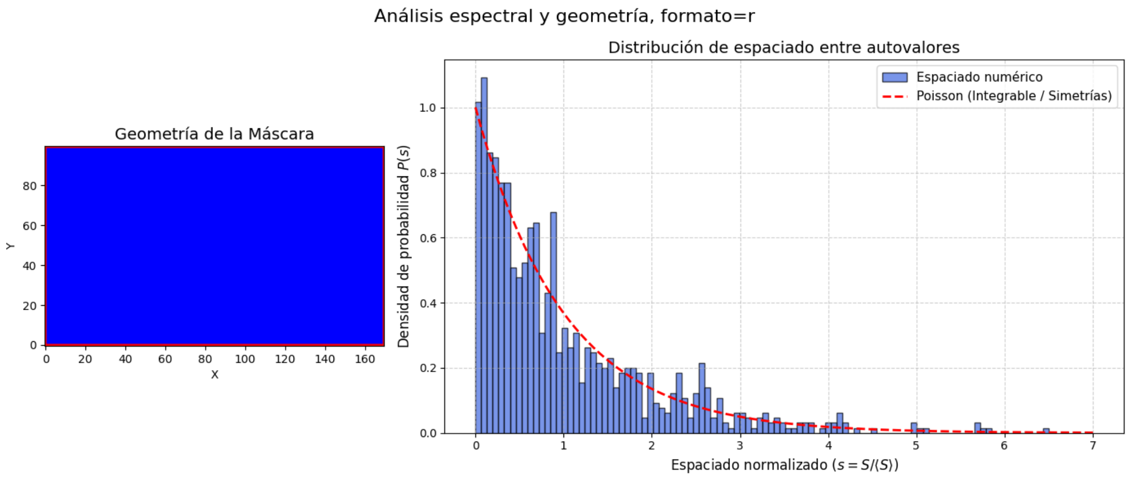


Figura 2: Poisson distribution for the integrable rectangle.

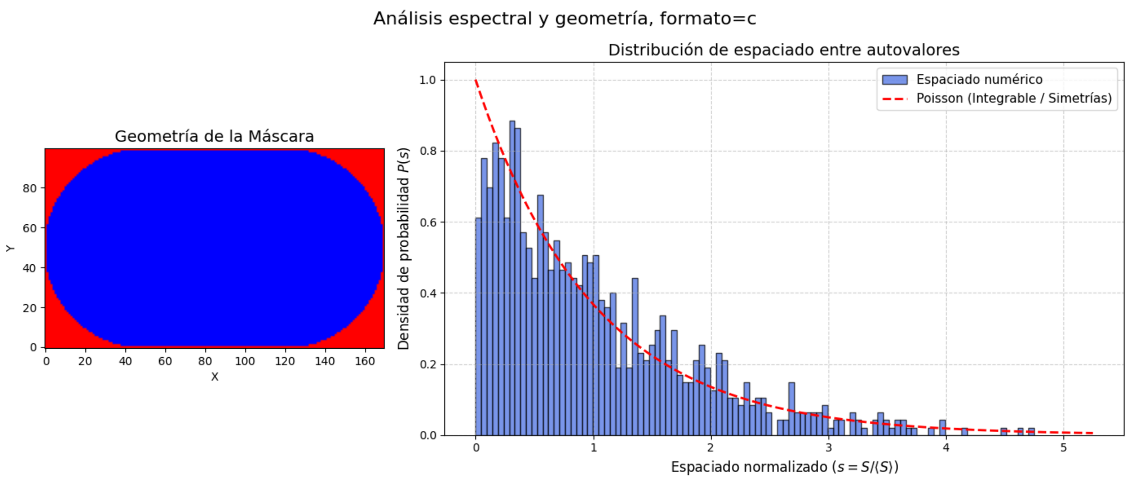


Figura 3: Poisson distribution for the non-integrable complete stadium.

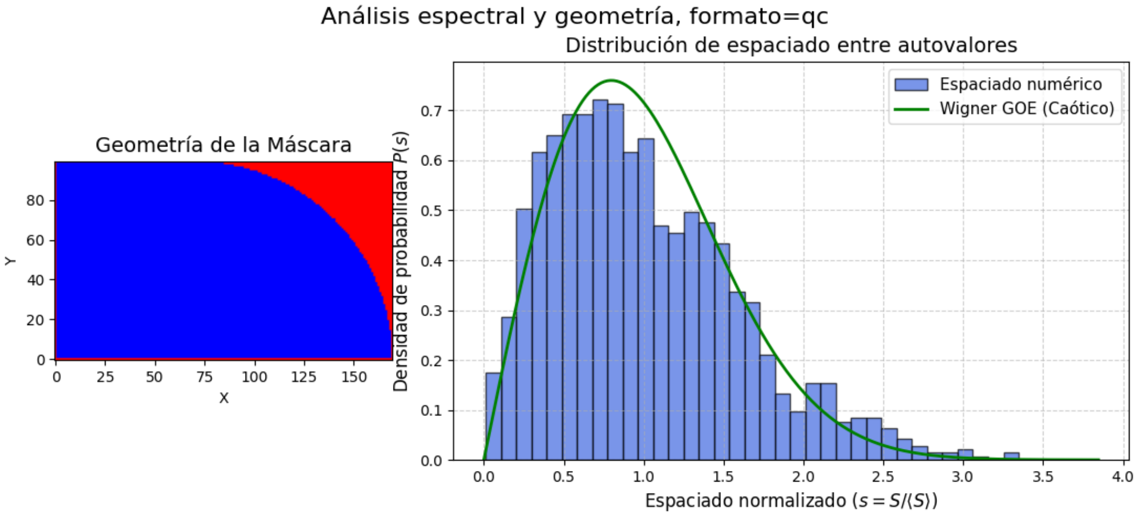


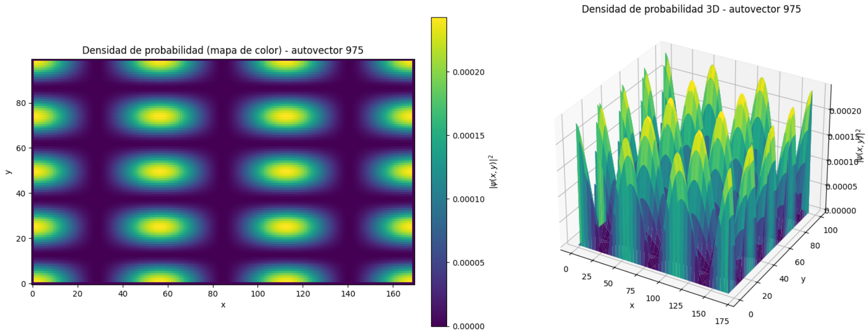
Figura 4: Wigner-Dyson distribution for a quarter of the non-integrable stadium.

As expected, we can clearly see in Figure 2 that the spacing distribution for the integrable rectangle follows a Poisson distribution. This graphic shows how this systems, even if it is integrable, that their eigenvalues are not correlated, then the probability of finding two eigenvalues close to each other is high ($P(s) \to 1$ when $s \to 0$). Also, in the case of the non-integrable billiad (the complete stadium showed in Figure 3), we can see that it also follows a near Poisson distribution, due to the symmetries of the system. This symmetries make the systm to have four pairs of solutions, then, the degenerate levels will increase the probability of finding two eigenvalues close to each other, even if the system is non-integrable.

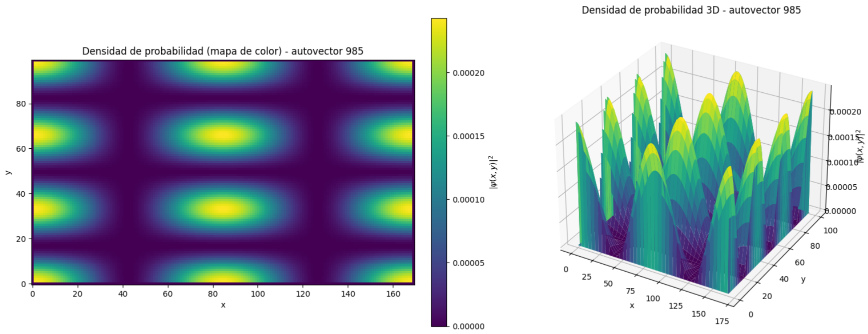
On the other hand, if we eliminate the $x - axis$ and $y - axis$ simmetries, having just a quarter of the stadium as we see in Figure 4, it is noticeable that the spacing distribution for the non-integrable billiard follows a Wigner-Dyson distribution. In this case, the opposite happens: the system may seem chaotic, but its eigenvalues are correlated, in a way that makes them repel each other or, in other words, the probability of finding two near-degenerate eigenvalues is very low ($P(s) \to 0$ when $s \to 0$).

3.2 High-energy states visualization

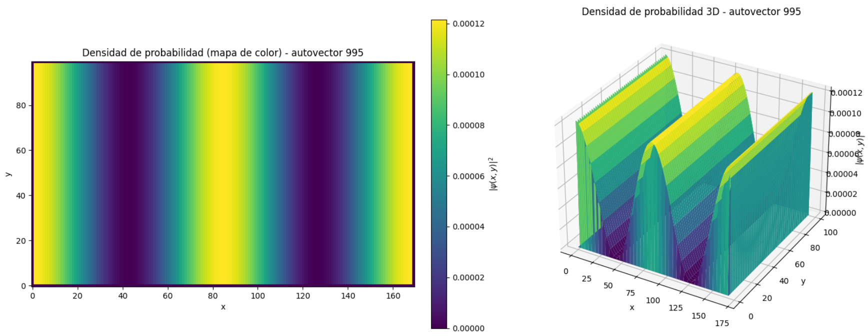
To determine the chaocity of the system, we must look at high-energy states, and compare them to the non-chaotic system. This is due to the fact that, in the low-energy states, the eigenvectors may seem like those of a string, therefore, they don't contain the necessary information to see if the system is chaotic or not. We then plot the probability density of some of the high-energy states for both systems.



(a) Probability density for the eigenvalue 975.

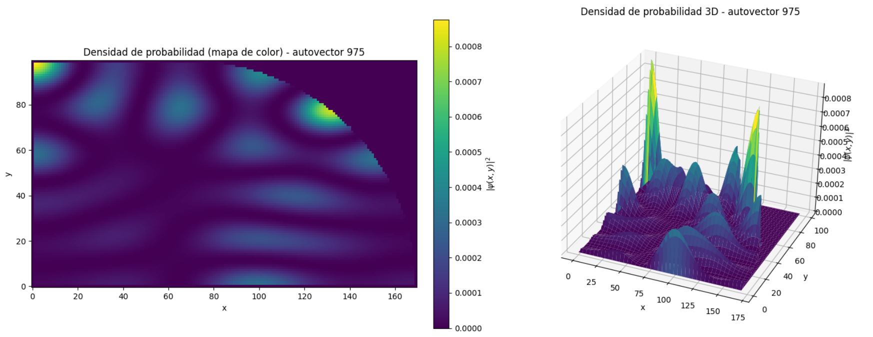


(b) Probability density for the eigenvalue 985.

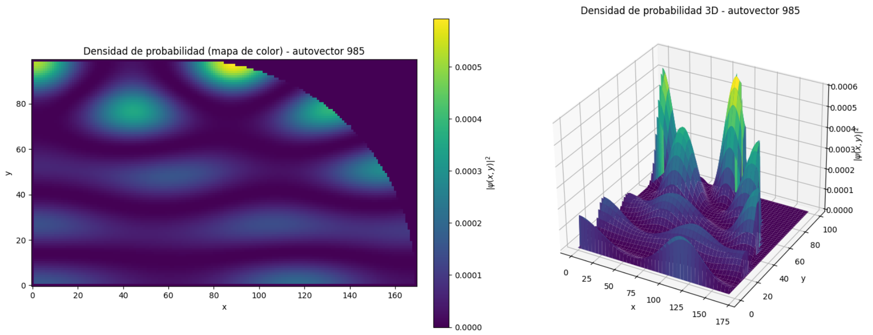


(c) Probability density for the eigenvalue 995.

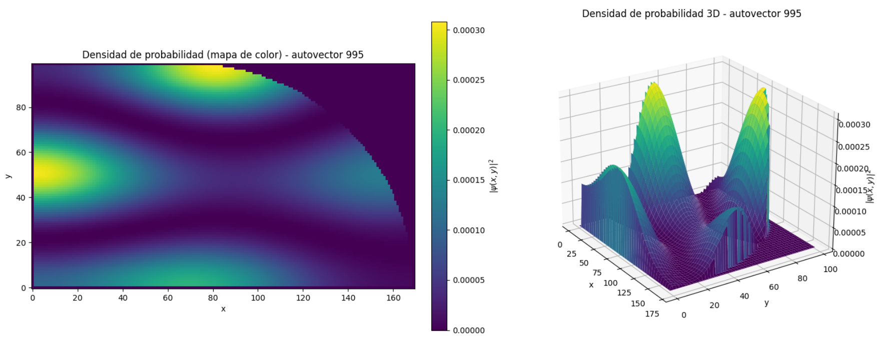
Figura 5: Visualization of various of the high-energy states for the **integrable rectangle**. The probability density is represented in a color scale, where the red areas correspond to higher probability density and the blue areas correspond to lower probability density.



(a) Probability density for the eigenvalue 975.



(b) Probability density for the eigenvalue 985.



(c) Probability density for the eigenvalue 995.

Figura 6: Visualization of various of the high-energy states for the **non-integrable quarter billiard**. The probability density is represented in a color scale, where the red areas correspond to higher probability density and the blue areas correspond to lower probability density.

The three graphics of Figure 5 show the probability density for three of the high-energy states of the integrable rectangle.We see that the patterns are very regular, and they look like the fundaental modes of a string, with nodes and antinodes. We can determine then that this system is not chaotic, as expected, since there are periodic patterns in the probability density.

In contrast, the same eigenvalues are shown in Figure 6. Is important to note that we only represented a quarter of stadium, because is the significant case in our study. In this system, the probability density shows 'quantum scars' or trajectories, wich corresponds to classical periodic orbits that a particle would follow within the stadium. Instead of a uniform distribution, these scars represent a quantum localization along specific geometric paths.

3.2.1 Ergodicity in the quantum system

In a classical system, a system is said to be ergodic when, after evolving for a sufficiently long time, a trajectory explores the accessible region of phase space uniformly. According to Shnirelman's theorem (also known as the quantum ergodicity theorem), this classical ergodicity is reflected in the corresponding quantum system: for sufficiently high-energy eigenstates, the probability density tends to become uniformly distributed over the accessible domain.

Since the original classical system is ergodic, the quantum system is expected to exhibit the properties described by this theorem:

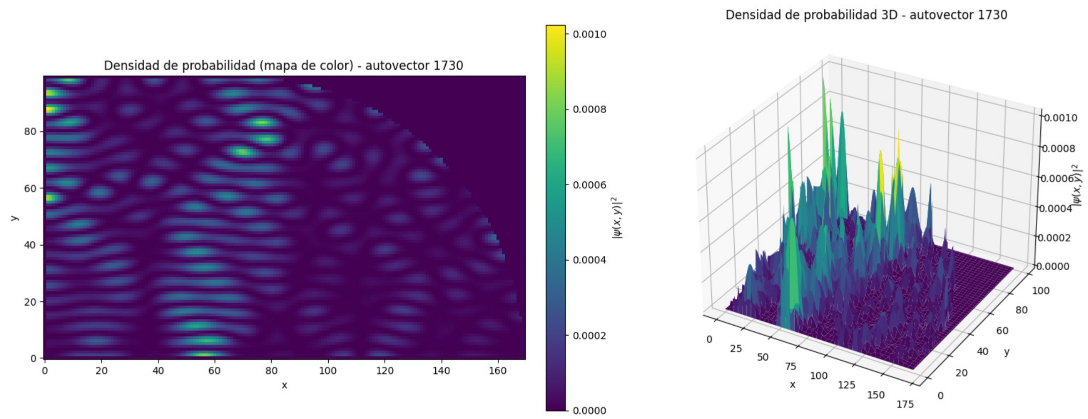


Figura 7: Probability density represented using a color scale, where red regions correspond to higher probability density and blue regions correspond to lower probability density. The figure shows an approximately uniform distribution over the accessible domain.

As shown in Figure 7, the probability density is distributed approximately uniformly throughout the domain. This behaviour is consistent with the predictions of the quantum ergodicity theorem for high-energy eigenstates.

4 Conclusions

The finite difference method combined with sparse matrices to perform the diagonalization results in a powerful technique for solving the 2D Schrödinger equation in arbitrary geometries. Using this approach, we obtain the eigenvalue spacings and observe signatures of chaotic behavior.

We show that the level spacing distribution in the rectangular billiard is Poissonian, which can be interpreted as the eigenvalues being uncorrelated. As a consequence, there is a high probability of finding two eigenvalues very close to each other or even degenerate. On the other hand, the spacing distribution in the stadium billiard follows Wigner-Dyson statistics, which indicates that the eigenvalues are correlated. Therefore, the probability of finding two very close or degenerate eigenvalues is strongly suppressed.

It is important to note that the complete stadium billiard possesses spatial symmetries (even and odd parity), causing the spectrum to become a superposition of independent subsystems. By considering the quarter-stadium geometry, these symmetry sectors are separated, allowing the pure Wigner-Dyson statistics to emerge without contamination from states of different parity. This isolation of the spatial symmetries is essential for obtaining undistorted spectral statistics that properly reflect the chaotic nature of the system.

In summary, this work successfully demonstrates how the complex statistical properties predicted by Random Matrix Theory naturally emerge when numerically solving a simple physical system under specific geometric boundary conditions.

Referencias

[1] Diego Ariel Wisniacki. *Caos cuántico en sistemas dependientes de un parámetro*. PhD thesis, Universidad de Buenos Aires. Facultad de Ciencias Exactas y Naturales, 1999.